

From Tensors to Agents

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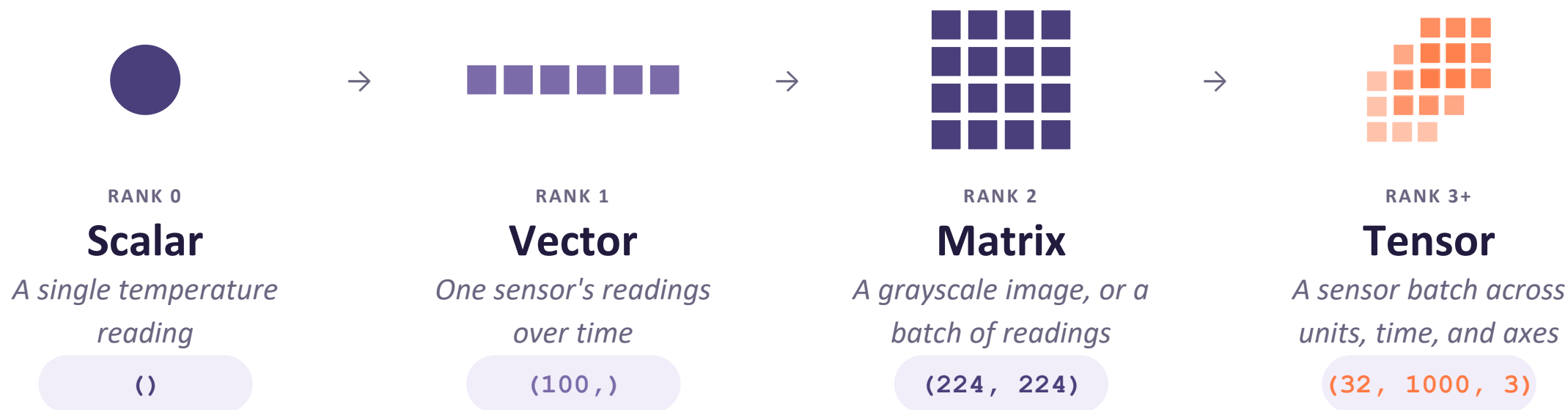
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Roadmap

- We open the black box behind
predictive (insight) → generative (content) → agentic (action)
- Predictive and Generative workflows
Text / Images → Tensors → Models → Prediction
Text → Tensors → Models → Text / Image Generation
- Getting to an Agentic Pipeline
Generation → Reasoning → Actions

What is a Tensor?

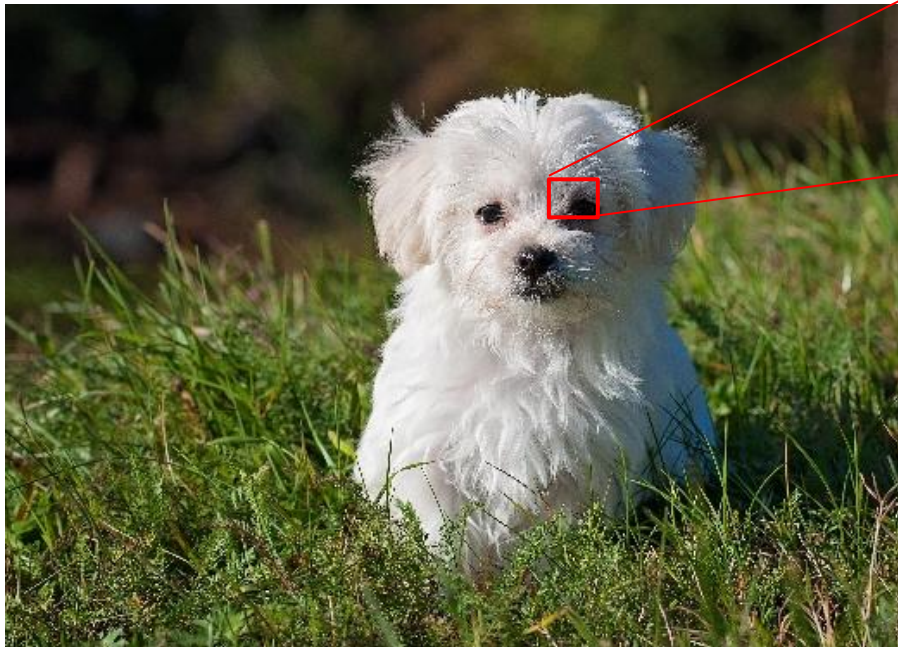
- A multi-dimensional (n-dimensional) array of numbers. Think of the generalization of a vector and a matrix.



- Every step in an AI/ML workflow uses tensor operations under the hood.

From Images to Tensors

Images are arrays of pixels value. All that the computer “sees” is a list of those numbers.



Shape = (8,12,3)

Predictions could be class predictions:

$Cat \rightarrow [1 \ 0 \ 0]$; $Dog \rightarrow [0 \ 1 \ 0]$; $Bird \rightarrow [0 \ 0 \ 1]$

Object detections:

$Dog \rightarrow [x_0, y_0, x_1, y_1, c]$

Image Segmentation

Content Generation

From Text to Tensors



Tokenization

Text is broken into subword units (tokens), each mapped to an integer ID — not quite words, not quite characters.

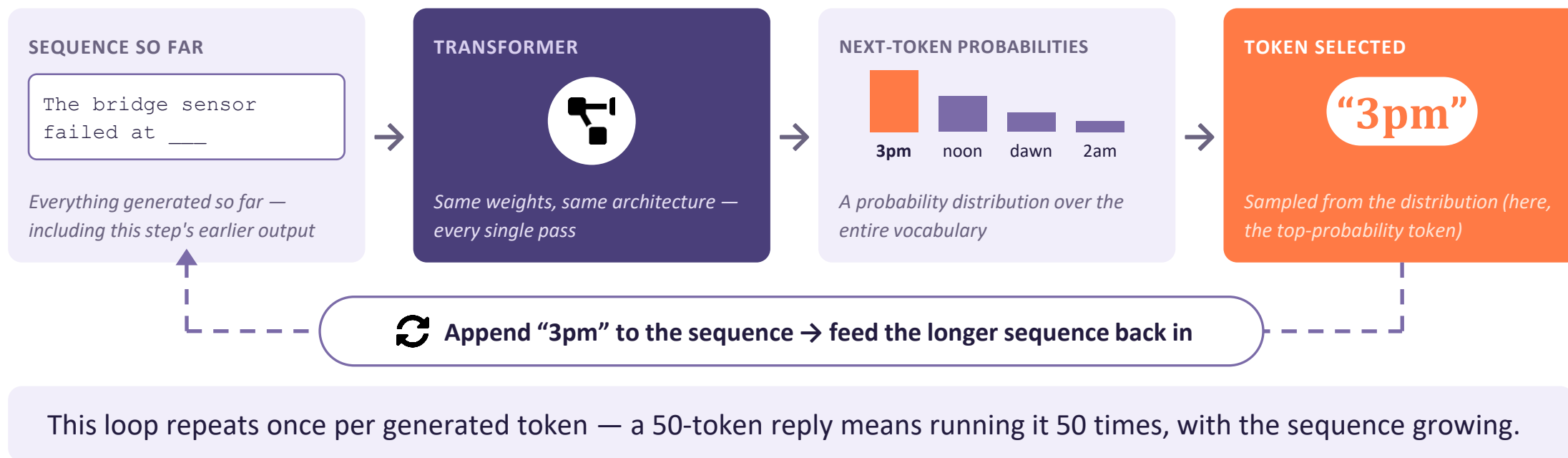
Embeddings

Each token ID is mapped to a vector in high-dimensional space, positioned so related meanings land near each other.

- Sequences of vectors are then used as input to models that can predict class labels, sequence labeling, etc.

Generating Text

- Each new token is predicted, appended and fed back in.



- The transformer itself is stateless – all memory lives in the growing sequence fed back in real time. The **context-window** is the input to the transformer. It has to be bounded.

From Images to Tensors (Revisited)

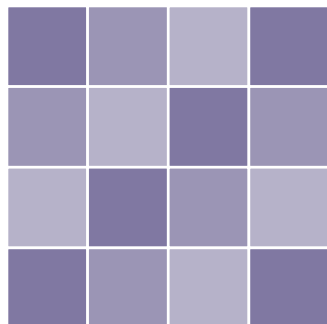
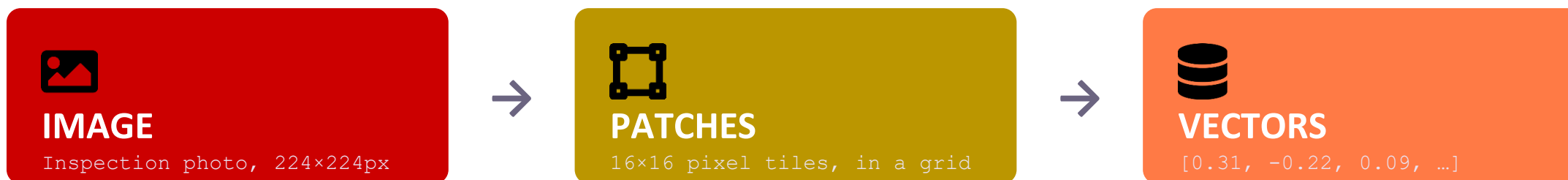


Image divided into a grid of patches

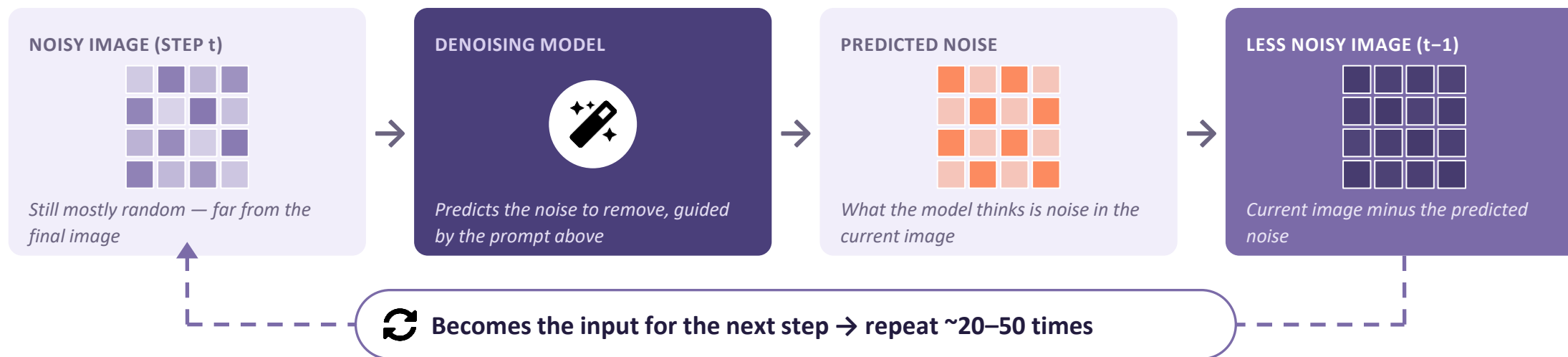
Each patch (or, in older CNN approaches, each learned filter response) becomes a vector — the same embedding-space idea from text, just applied to pixels instead of tokens.

- Input images don't need to be processed sequentially over space.

Generating Images

- Each pass removes a little noise and feeds the results back in

A TEXT PROMPT: “a reinforced concrete bridge at sunset” — the same prompt conditions every single step below



This loop repeats once per denoising step — a typical image takes 20–50 steps.

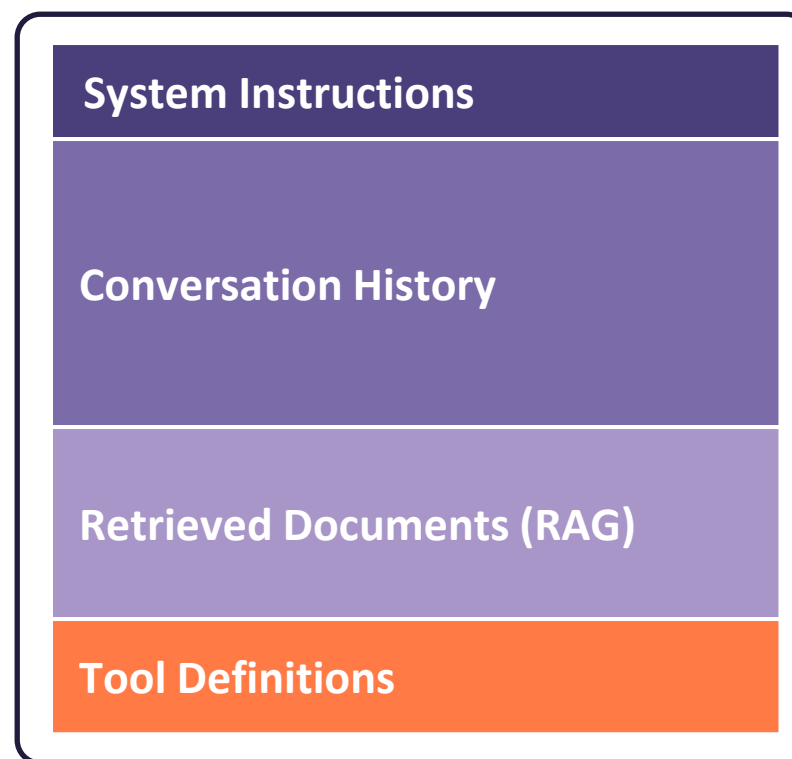
- The denoising model is stateless – all memory lives in the image tensor passed between steps.

From Generative Model to Assistant

Context: The Model's Working Memory

- Everything the model “knows” right now lives in this one limited space
- It has no persistent memory between calls unless something explicitly puts information back into context
- Context Engineering decides what goes into this limited space, and in what order
- Once something falls outside the window, the model has no mechanism to say: “I don’t have that”. It just continues generating plausible content.

CONTEXT WINDOW



Personas: Shaping Behavior, Not Knowledge

- The same question, the same model, two very different answers



“Patient Tutor”

Q: Why isn't my gradient descent converging?

Great question — let's work through it. First, check your learning rate: if it's too high, updates can overshoot the minimum and bounce around instead of settling. Try reducing it by 10x and see what happens...



“Terse Code Reviewer”

Q: Why isn't my gradient descent converging?

Learning rate too high. Reduce by 10x. Also check for unnormalized features — likely culprit given your loss curve.

Hallucinations: Confident, Not Necessarily Correct

- The model predicts plausible text. It has no built-in “I don’t know” signal.



Model's apparent confidence

Consistently high — every answer sounds equally certain

Actual factual accuracy

Varies — depends on training data and topic

- The model is producing the statistically plausible next token, not consulting a fact database. When it doesn't “know” something, it just continues the pattern rather than stopping.

RAG vs. Fine-Tuning

- RAG (Retrieval-Augmented Generation) is really an automated way of deciding what goes into Context — based on relevance, not fixed in advance.

	RAG	Fine-Tuning
What changes	Context, at request time	The model's weights
Update speed	Instant — swap the source	Requires retraining
Source attribution	Yes — traceable to a retrieved doc	No — baked into weights
Best for	Fast-changing or large knowledge bases	Stable shifts in behavior or style

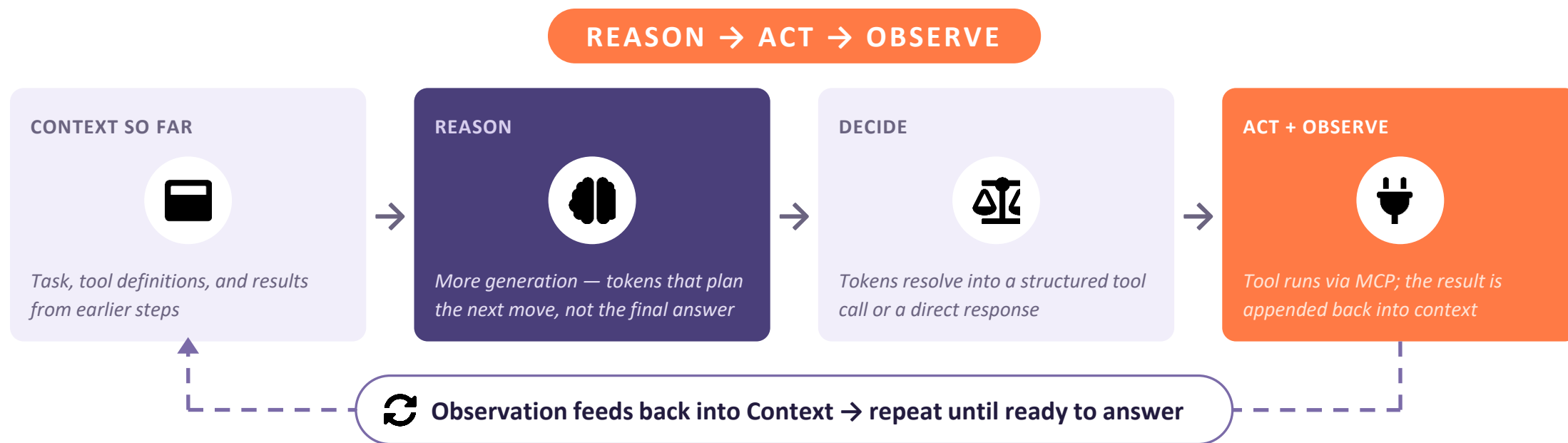
From Generative Models to Assistants

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From Generation to Action

Reasoning

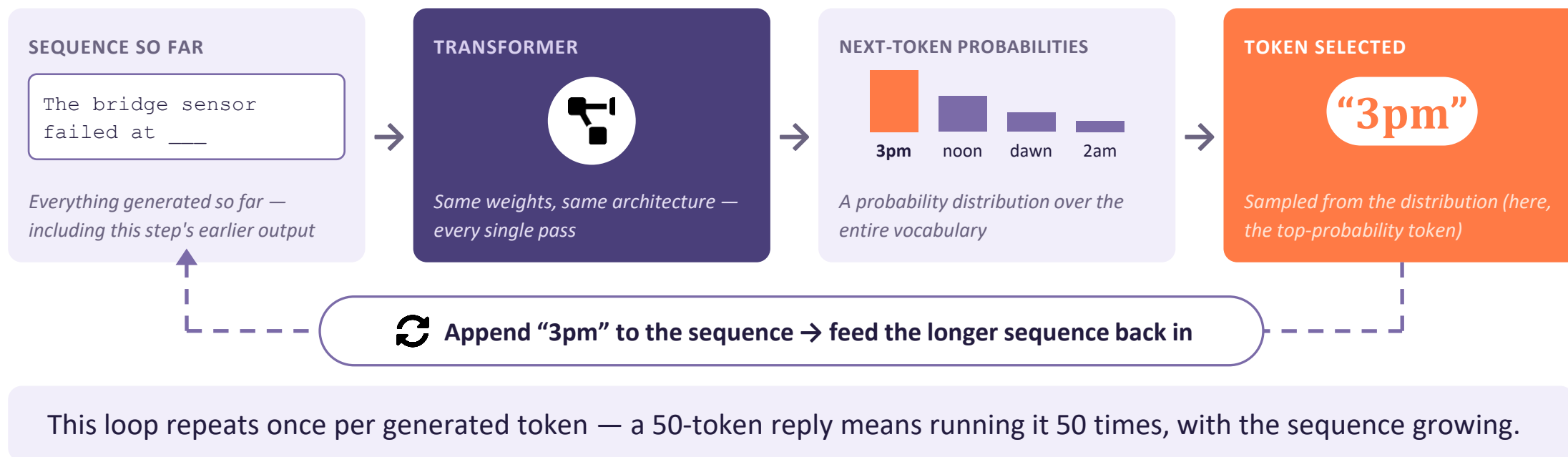


This loop can run many times — each pass may call a different tool, refine the plan, or decide it's ready to answer.

- Does this look familiar?

Generating Text

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Skills: Packaged Procedural Expertise

- Reusable knowledge the model reads before doing a specific class of tasks



Instructions, examples, templates — read before the task, not baked into weights

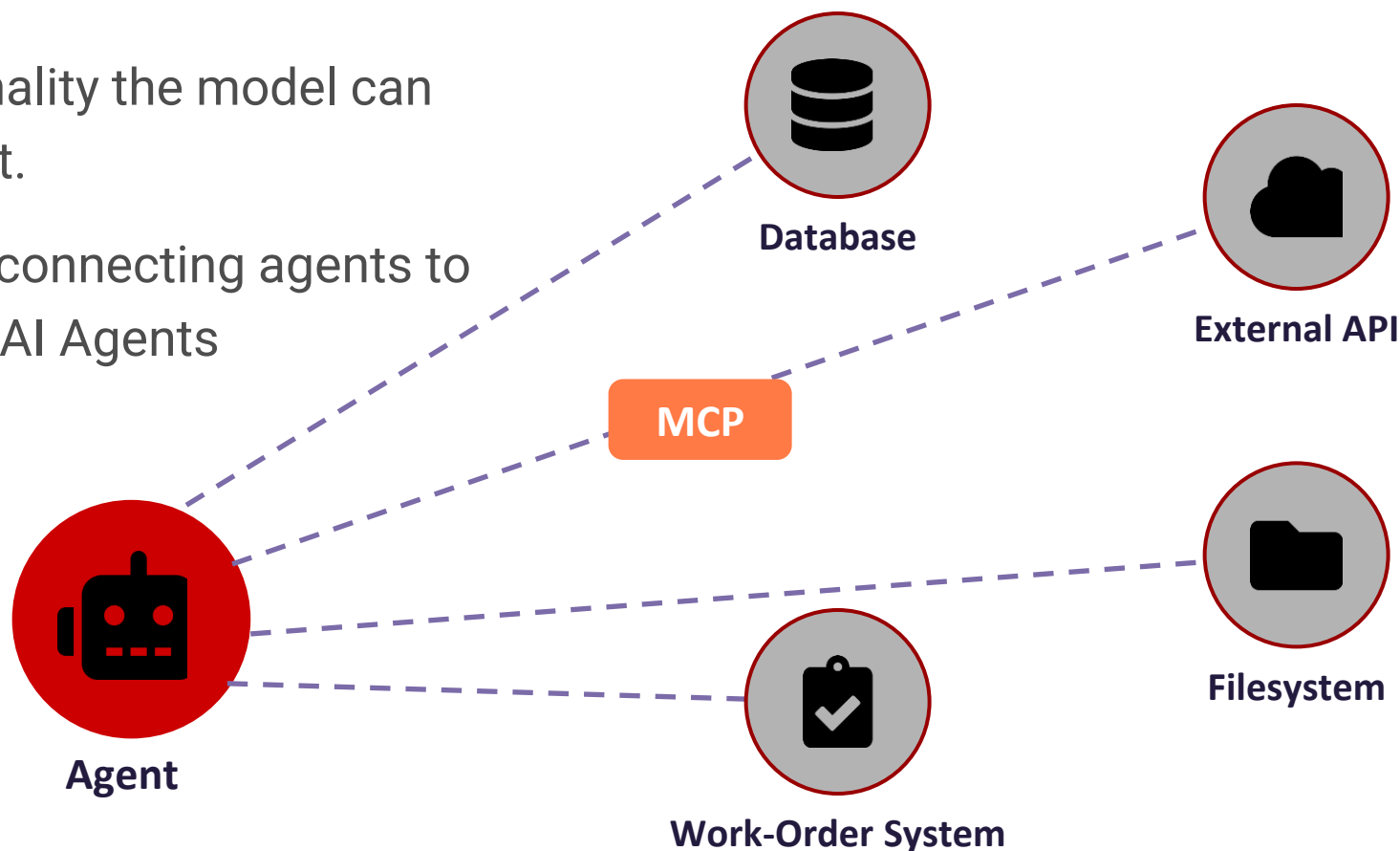
Engineering analogy: a skill is functionally a standard operating procedure (SOP). A specialist doesn't re-derive the correct inspection procedure from first principles each time — they follow a written procedure refined over many iterations.

This is now a first-class concept in production agent platforms — not just an informal prompting trick.

- While Personas set the role, Skills supply the know-how for a specific tasks within that role

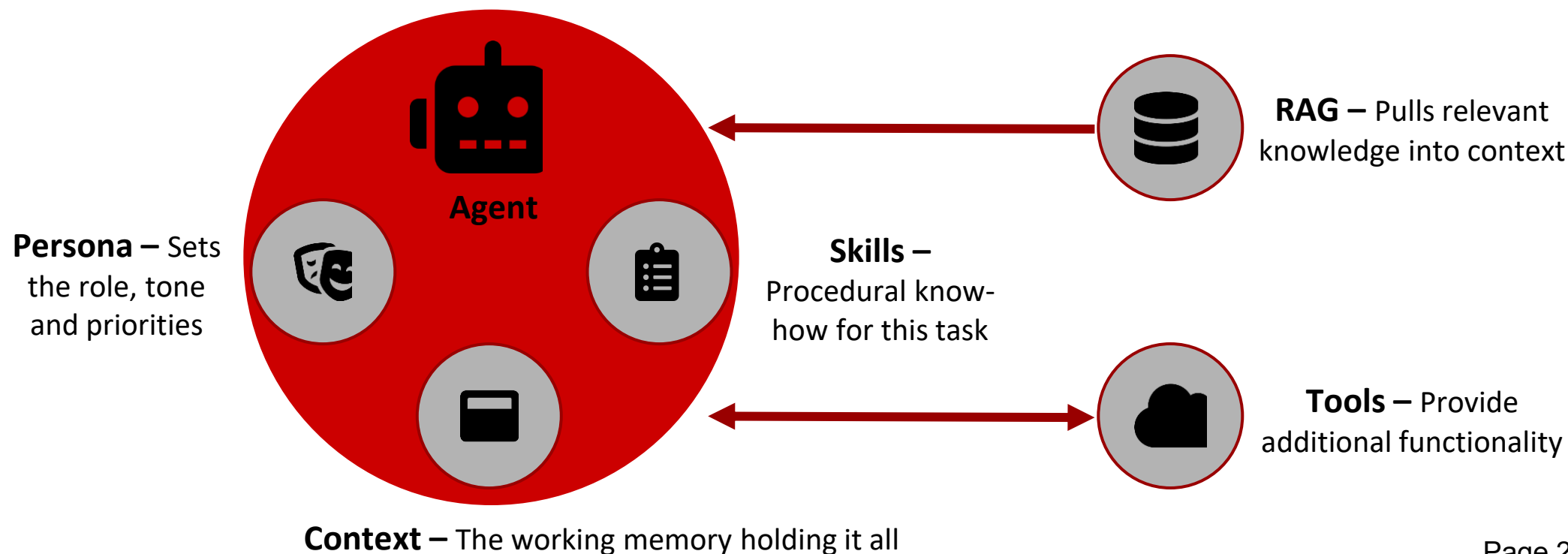
MCP: Model Context Protocol

- A tool is any external functionality the model can invoke beyond generating text.
- MCP is an open standard for connecting agents to tools and data, “the USB-C of AI Agents”



Putting It All Together

- An agent incorporates a persona, with skills, calling tools via MCP, grounded by RAG, making repeated generative calls in a loop.



From Generation to Action

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